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Circularity Measurement and Management

A Framework for Evaluating and Enabling Circularity in a Micromobility Product-as-a-Service System

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Abstract

As circular economy (CE) principles gain traction and reporting requirements tighten, Product-as-a-Service (PaaS) companies face increasing pressure to measure and manage circularity effectively. This thesis develops a circularity measurement framework tailored to a micromobility PaaS context by evaluating five CE models—CPDM2, DfPC, FOUT, CSFP, and PLCM—against the ISO 59020 circularity measurement standards and business relevance criteria. The framework combines core ISO indicators with custom metrics, including Time-to-Repair and Utilisation Rate, to better reflect the operational dynamics of micromobility PaaS products. Applied to a leading micromobility operator, results show that improving repairability, extending product lifespan, and increasing utilisation have greater impact than focusing on recycled content alone. The study also reveals gaps in circular design maturity and measurement practices, particularly in early product development. By integrating targeted indicators and promoting modular, repair-friendly design, companies can better align CE strategies with business goals. In addition to enabling measurement, the framework also supports circularity management by informing KPI development and guiding implementation efforts. The proposed framework offers practical guidance for enabling circularity in PaaS companies.

Keywords: Circular economy, Measuring Circularity, Sustainable design, Slowing Closing Resource loops, Design for Sustainability, Micromobility, R Framework, Product-Service System, Indicators

Sammanfattning

I takt med att principerna för cirkulär ekonomi (CE) får allt större genomslag och rapporteringskraven skärps, ökar pressen på företag som erbjuder Product-as-a-Service (PaaS) att på ett effektivt sätt mäta och hantera cirkularitet. Denna uppsats utvecklar ett ramverk för cirkularitetsmätning anpassat till mikromobilitet inom PaaS, genom att utvärdera fem etablerade CE-modeller—CPDM2, DfPC, FOUT, CSFP och PLCM—mot ISO-standarden 59020 samt mot affärsrelevanta kriterier. Ramverket kombinerar nyckelindikatorer från ISO med anpassade mått, såsom reparationstid och nyttjandegrad, för att bättre spegla de operativa förutsättningarna för mikromobilitetsprodukter i en PaaS-kontext. Vid tillämpning på en mikromobilitetsoperatör visar resultaten att förbättrad reparerbarhet, längre livslängd och ökad nyttjandegrad har större effekt än att enbart fokusera på andelen återvunnet material. Studien identifierar även brister i både mognadsgrad av cirkulär design och mätmetoder, särskilt i de tidiga faserna av produktutvecklingen. Genom att integrera riktade indikatorer och främja modulär, reparationsvänlig design kan företag bättre anpassa sina CE-strategier till sina affärsmål. Utöver att möjliggöra mätning stödjer ramverket även cirkulär styrning genom att bidra till utvecklingen av KPI:er och vägleda insatser för implementering. Det rekommenderade ramverket erbjuder praktisk vägledning för att möjliggöra och stärka cirkularitet inom PaaS-baserade affärsmodeller.

Nyckelord: Cirkulär ekonomi, Mätning av cirkularitet, Hållbar design, Slutning och förlängning av resursflöden, Design för hållbarhet, Mikromobilitet, R-ramverk, Produkt-tjänstesystem, Indikatorer

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1 Introduction

This first chapter consists of a short introduction and background to the subject matter and will then be followed by a purpose statement and formulation of the research questions.

1.1 Background & Problem Statement

There is a growing need for companies to clearly measure all activities that affect their carbon footprint and influence their broader sustainability performance. This demand is driven by new legislation and regulatory guidelines that goes into effect for countries in the European Union within the next couple of years. As of 2024 [1], businesses will be required to report in accordance with the European Sustainability Reporting Standards (ESRS), including ESRS E5 on resource use and the circular economy (CE), to comply with the Corporate Sustainability Reporting Directive (CSRD) (Ellen MacArthur Foundation 2023; *Corporate sustainability reporting - European Commission* 2025). Of particular relevance is ESRS E5, which requires organisations to disclose how they manage resource use and CE strategies. Unlike traditional sustainability metrics focused on emissions or energy, circularity requires new indicators to track material loops, product lifetime, reuse, repairability and waste minimisation.

These regulatory shifts creates pressure on organisations to not only reduce impact, but to quantify circular strategies and link them to business outcomes (Michel Bajuk et al. 2024). In the Circularity Outlook report by Bajuk and Linder it's clear that a majority of the companies in the study have CE as a part of their business strategy, however it showed that only 40% of the companies had formulated at least one circularity target. This reflects a significant gap between intentions and actionable metrics, pointing to a need that go beyond carbon calculations. Such frameworks enable companies to comply with ESRS, make informed design and operational decisions, and unlock the full value of circular transitions.

Micromobility operators handle fleets of smaller, lightweight vehicles (generally e-scooters or e-bikes) mainly used for transportation over shorter distances. Either powered by humans or smaller electric motors. The business models of these operators can be described as a Product-as-a-Service model. Which can simply be described as instead of buying the products, customers pay for a service that that allows the customer to use the product for a period of time.

Within Europe, there is a handful of different actors that use procurements to try to win cities in which they want to operate. The micromobility industry is therefore very competitive and demands quick advancements and improvements of the vehicles to stay up to date and ahead of competitors.

The project sponsor - also called the case company in this thesis - is one of these actors. The case company wants to play an important role when it comes to advancing sustainable urban transportation which in turn contributes to achieving sustainability goals. The case company aims to be the first fully circular micromobility solution and seeks to explore a structured approach to achieve this.

While there is a growing body of research on CE principles and the development of methods for measuring circularity, studies focused specifically on product- and company-level circularity remain a relatively unexplored context - particularly in evolving sectors

[1] During the course of this thesis, this has been postponed one year.

like micromobility. Research appears especially limited in how CE design principles can be systematically integrated into business models while also accounting for financial viability (Mesa et al. 2023). These gaps are even more apparent in sectors characterised by fast-paced product lifecycles and unique challenges related to maintenance, material recovery, and consumer perceptions of recycled components.

1.2 Purpose

In light of the challenges described in the previous section, this thesis seeks to draw upon emerging frameworks in circular product design, lifecycle thinking, and sustainability reporting. The aim is to apply academic circularity frameworks to real-world data and operational constraints in order to develop measurable, business-relevant indicators that improve circular performance in micromobility systems.

The purpose of this master thesis is to develop a measurement model for circularity and build a business case for the case company's circular micromobility system. This will include considerations of different aspects such as cost implications, carbon impact and the potential for extending product lifetime. The evaluation of the business case will be supported by a conceptual design of a sustainable e-vehicle which will have its focus on maximising the use of recycled materials, optimising repair time and be designed for reuse. Primarily, the conceptual design will serve as a case study in order to analyse critical aspects of circularity.

The business case developed in this project should indicate the effects on business and carbon emissions of the analysed circular options as well as providing guidance on business viability of the alternatives.

The research questions are as follows.

- RQ1: In what ways can existing circular measurement models be applied for a micromobility Product-as-a-Service company?
- RQ2: How does increasing the recycled content of critical components within a micromobility product affect the circularity and carbon emissions of a Product-as-a-Service company?

2 Theoretical Framework

This section will provide a theoretical background to the subject of CE, strategies and circular business models. It will then continue to provide a definition of CE indicators and describe the measurement models investigated in this thesis.

2.1 Defining Circularity

2.1.1 Circular Economy and Circular Strategies

One of the most well-known definitions of CE was framed by the Ellen MacArthur Foundation (2013, p. 7) that describes it as *"... an industrial system that is restorative or regenerative by intention and design. It replaces the 'end-of-life' concept with restoration [...] and aims for the elimination of waste through the superior design of materials, products, systems, and within this, business models."*. The foundation highlights that CE aims to replace the traditional linear economy's "end-of-life" concept with a model that prioritises restoration and waste elimination through superior design of materials, products, and business models.

A key aspect of defining circularity is distinguishing between different types of circular loops. Potting et al. (2017) classifies circular strategies into different levels ranked based on their circular potential from high to low. The R's with lower numbers, such as R1 Rethink or R2 Reduce, has a higher focus on smarter product use and manufacture by e.g. sharing the products or increasing efficiency by using less natural resources. The higher R's are strategies aimed at extending product lifespan, or handling material either by recycling or recovering from discarded products, see Table 1 for the different R strategies explained more in detail.

Table 1: List of the R-strategies

Strategy	Description
R0 - Refuse	Avoid the use of products or materials by eliminating unnecessary functions or by replacing them with radically different solutions.
R1 - Rethink	Make usage of the product more intensive and efficient, (e.g sharing the product)
R2 - Reduce	Increase the efficiency of product manufacture or use by minimising the consumption of natural resources and materials.
R3 - Reuse	Extend the product lifespan by using discarded products that are still in good condition over again.
R4 - Repair	Maintain and repair defect products so that it can be used for its original function
R5 - Refurbish	Restore and update old products or components to ensure that they remain functional and relevant.
R6 - Remanufacture	Use parts of discarded product in a new product with the same function
R7 - Repurpose	Use parts of discarded product in a new product with a different function
R8 - Recycle	Process materials to obtain the same or lower quality
R9 - Recover	Extract energy from waste materials when further material reuse or recycling is not possible.

The R framework can be a helpful tool to assess circular strategies, not only when it comes to material flows, but also highlight the systemic changes that are required for more ambitious levels of circularity.

Bocken et. al (2016) introduced a cyclical approach for the development of products and systems which distinguishes from traditional linear models by designing business model

strategies that are slowing, closing and narrowing resource loops. These strategies aim to reduce resource extraction, minimise waste and extend the utilisation period of products and materials over time. Slowing loops includes extending the use and reuse of products over time by designing the products for maintenance and repair, upgradability, standardisation, compatibility and disassembly. Closing resource loops, on the other hand, is about the reuse of materials through recycling. This strategy includes developing products that enables the materials to continuously and safely be recycled into new materials and products. The aim here is to keep the value of the materials and continually cycle them within the economy. Narrowing resource loops however, includes reducing resource use within the product and production process but it does not influence the speed of flow and does not involve service loops such as repair and maintenance. Excluding the time dimension can sometimes lead to further speeding up of linear resource flows like selling more of a more efficient product which in turn can result in small overall savings. Although this third approach is focused on minimising resource input, it differs from closing and slowing the loop as it does not change how long the product is used or if the materials are recycled.

Although circularity and sustainability are clearly related, they are not synonymous. Sustainability encompasses a wide range of environmental, social and economic factors that include the mitigation of climate change, the protection of biodiversity, and social sustainability. Circularity however, as Geissdoerfer et al. (2017) claimed, can be viewed as a condition for sustainability and prioritises the economic system with environmental benefits and implicit social gains. While sustainability can be viewed as an "open-ended" system with broad objectives that can vary by context, circularity is aimed at creating closed-loop systems that eliminate resource inputs and waste.

Given this, it becomes essential to establish clear methods for evaluating the effectiveness through specific indicators to track the progress in resource efficiency, waste reduction, and material recovery.

2.1.2 Circular Business models

Reducing the dominance of product ownership is one way to increase the level of circulation of products. Product-as-a-service (PaaS) or usage-focused business models provide access to a product rather than selling it. Instead of owning, the customer pays for using the product, usually through a subscription, rental, or pay-per-use model. PaaS contributes to circularity by shifting the focus to long-term sustainability, reuse, and efficiency and optimises resource use, extends product life cycles, and encourages sustainable business models (Tukker 2015). Furthermore, PaaS systems mean that companies retain ownership and therefore have stronger incentives to design products that are more durable, maintainable, and upgradable (Bressanelli et al. 2018). By ensuring longer product life cycles, reducing waste, and enabling take-back systems, these models directly contribute to the goals of CE.

However, PaaS systems are not a guarantee for resource reduction, nor absolute resource decoupling [2] (UNEP 2011; Kjaer et al. 2019). A risk when measuring the effectiveness of CE strategies is to focus on optimising the end-of-life stage, at the expense of other stages in the product life, like production.

Zink and Geyer (2017) argues that CE principles do not automatically lead to environmental benefits because of CE rebound - the phenomenon where secondary production (reuse,

[2]Resource Decoupling: reducing the link between economic growth and the consumption of natural resources

refurbishment and recycling) increases total production and consumption rather than displacing primary production. Although PaaS systems are pillars of CE, they come with specific rebound risks. PaaS can lower the barriers to product access and use, which can unintentionally drive higher consumption. For example, peer-to-peer sharing of vehicles or equipment means more people can afford use of the product, potentially increasing the total user base or usage rate. A study of an online boat-sharing platform showed that while sharing improves utilisation of idle assets, it also enabled greater overall boat usage and related emissions; in fact, about 30% of participants ended up with higher carbon footprints than if they hadn't participated (Warmington-Lundström et al. 2020). This kind of backfire often happens in PaaS when new usage augments rather than replaces existing consumption.

Another rebound pathway in PaaS is via cost savings. If renting a product is cheaper in the short term than buying it, users may spend the freed-up money (or time) on additional consumption. The sharing economy and PaaS models carry a dual rebound risk – on the consumer side (due to increased access and lower costs) and on the provider side (due to income effects). Empirical studies have documented that ride-sharing, car-sharing, and accommodation-sharing can lead to additional trips or resource use that diminish their sustainability gains (Siderius et al. 2021).

Because rebound effects can jeopardise the intended environmental gains of a CE, researchers and policymakers have been exploring strategies to anticipate and mitigate them. Identifying and measuring rebound effects would be a first step. Tools such as elasticity analysis which estimates how changes in efficiency or prices affect demand; econometric models that use statistical techniques to identify relationships between variables over time; input-output models that track how changes in one sector affect others across the economy; and system dynamics models that simulate complex interactions and feedback loops in systems over time can be used to estimate rebound in circular contexts (Lowe et al. 2024). The rebound effects highlight the importance of critically evaluating circular strategies beyond their theoretical benefits. Understanding potential rebound effects is necessary for selecting appropriate circularity indicators and ensuring that measurement frameworks account for both intended and unintended outcomes.

In addition to technical and economic rebound risks, research has drawn attention to **aesthetic obsolescence** — a phenomenon where users replace functioning products due to visual wear, design trends, or branding rather than performance (Sierra-Fontalvo et al. 2024; Alzaydi 2024). This form of non-functional obsolescence can undermine circular strategies, particularly in sectors like micromobility where product appearance plays a role in perceived value and competitiveness.

2.2 Measuring Circularity

2.2.1 Circularity Indicators

A circular indicator represents a quantitative or qualitative measurement of a circularity aspect [3] and can be used at any cycle phase of a solution (ISO 59010 2024). These indicators are metrics used to assess the implementation and impact of CE strategies and help assess how resources are preserved and reused in a closed-loop system. However, due to the broad

[3]Circularity aspect: an element of an organisations activities or solutions that interacts with the circular economy

and evolving nature of CE, there has for a long time not been any universally agreed upon set of indicators and different approaches exist on three scales:

- **Micro level:** which focuses on products, materials and individual organisations
- **Meso level:** which looks at eco-industrial parks or B2B networks
- **Macro level:** which involves cities, regions, or entire nations (Elia et al. 2017).

During 2024 and 2025, the ISO 59000 family of standards have been developed to help facilitate the transition to CE by providing structured frameworks for measuring, implementation and adoption (*ISO - Circular economy* 2025).

Within these standards exists a definition of indicators as well as a set of mandatory indicators that should always be used when measuring circularity. These set of indicators from the ISO can be viewed in the appendix, A.1. The ISO 59020 standard presents the following five points that define an indicator:

- Circularity aspect
- System level
- Measurement unit
- Data Requirements
- Calculations or aggregation needed to convert the acquired data into a form or numeric value

The aspect of circularity they assess can include material flows, energy efficiency, land use, economic performance, and environmental impact. A key distinction between indicators are **direct indicators**, which measure specific circular actions such as recycling rates or resource efficiency, and **indirect indicators**, which should provide insight into broader systemic effects, such as economic resilience and social well-being (ISO 59010 2024). A distinction is also made between the **core indicators**, which is a set of quantitative indicators that should be considered for circularity measurement and assessment, and **additional indicators** which can supplement the core indicators to meet the intended circularity measurement scope. The minimum set of quantitative core circularity indicators that should be considered for circularity measurement and assessment can be viewed in A.1.

However, even with the development of the ISO standards, comprehensive methods for assessing CE across diverse sectors and levels remain lacking (Ferrante et al. 2025). Due to the way the ISO is built, beyond the mandatory indicators there still is room for interpretation and wide choices in choosing indicators. This means that there still lacks consistency, comparability and a clear way for companies to measure their CE efforts, especially when applying the ISO to complex systems (Macias Aragonés et al. 2024; Vinante et al. 2021; Kirchherr et al. 2017)

According to Moraga et al. (2019), CE indicators provide a structured approach to measuring circularity but require a combination of different types to capture the full picture. Moraga et al. mention several CE measurement models and frameworks that use indicators like Material Flow Analysis (MFA), which tracks physical flow of materials through a system, and Life Cycle Assessment (LCA), which evaluates the environmental impacts associated with all stages of a products lifecycle. Circularity Gap Analysis is another approach, assessing how

much of the economy is circular vs linear. However useful, these models do not alone give a complete measurement of circularity. There is an overemphasis on material usage and a need for models that evaluate product longevity, service-based circularity, and broader environmental and social impacts (Corona et al. 2019). Corona et al. notes that LCA is one of the most comprehensive tools to use because it accounts for multiple environmental impact categories but its application in CE needs to be complemented to avoid narrow one-dimensional evaluations.

Further, comparisons between LCA results and CE indicators for product-level measurement illustrate that they are not necessarily aligned (Jerome et al. 2022). Jerome et al. support the view that CE indicators often focus too heavily on materials and recycling. They further argue that many indicators are skewed, as they tend to neglect important aspects such as resource flows during the use phase, maintenance, and product repurposing. In addition, most indicators do not capture multiple use cycles, which limits insights into reuse and remanufacturing strategies. Jerome et al. therefore conclude that CE indicators cannot replace LCA and should instead be seen as complementary tools.

2.2.2 CE Measurement Models

To complement the ISO standards indicator framework, it could be insightful to also look at other CE assessment models that could more easily be visualised and connected to business strategies. In these following sections, five frameworks will be presented. An overview of these can be viewed in Table 2.

Table 2: List of CE assessment models in the order of which they are presented

Model Abbreviation	Model name	Year	Authors
PLCM	Product-Level Circularity Metric	2017	*Linder, M. Sarasini, S. Van Loon, P.*
CPDM2	Circular Product Design Maturity Matrix	2022	*Aguiar, M F. Jugend, D.*
CSFP	Closing-Slowing Future-Past Model	2022	*Shevchenko, T. Yannou, B. Saidani, M. et al.*
FOUT	Functionality Over Use-Time model	2022	*Hatzfeld, T. Backes, J G. Guenther, E. et al.*
DfPC	Design for Product Circularity model	2024	*Shevchenko, T. Cluzel, F. Yannou, B. et al.*

Product-Level Circularity Metric

Linder et. al (2017) proposed to fill the gap on metrics to quantify circularity at the product level. They argued that economic value, rather than mass or energy is the most appropriate unit to measure circularity, because it reflects relative scarcity, market dynamics and demand. Their Product-Level Circularity Metric (PLCM) is designed to be operational for businesses and can therefore offer valuable insights in how to use circularity measurements in business decisions. Their approach focuses on measuring only circularity, excluding factors like environmental impact or competitiveness, to maintain clarity. The key idea with PLCM is to quantify the fraction of the economic value of a product that comes from recirculated parts (e.g., reused, remanufactured or recycled components), in accordance to equation 1 below.

$$c = \frac{\text{economic value of recirculated parts}}{\text{economic value of all parts}} \tag{1}$$

Even though this method is designed to be scalable and practical across value chains, it

does however exclude product lifespan from the metric, which might undervalue long-lasting products.

Circular Product Design Maturity Matrix

Another developed model for measuring circularity is a framework called Circular Product Design Maturity Matrix, (CPDM2), with the purpose to gradually adopt CE into the New Product Development (NPD) process (Aguiar et al. 2022). The literature emphasises the importance of incorporating circular strategies early in the design process as many important decisions are made during these preliminary phases, such as material selection. While much of existing research has had its focus on circular business models, Aguiar et. al argue that less attention has been on the design processes.

The maturity matrix covers 11 dimensions related to both design strategies and design processes with the intention to map out different approaches for implementing circularity. Some of the dimensions align well with key CE strategies such as narrowing, slowing and closing the loop, while others have their focus on broader aspects including stakeholder collaboration and the use of lifecycle data. For clarity, narrowing the loop involves strategies like designing for modularity which allows for easier component replacement. Slowing the loop includes designing for repairability, durability, and fostering user trust and attachment that extends product lifespan. Closing the loop is focusing on using recycled materials, reducing the diversity of materials to promote recyclability and ensuring easier disassembly at end-of-life (EoL).

The CPDM2 covers five levels of circular maturity, from Inert which means no experience with circular product design, to Optimised where stakeholder collaborations is high and circular performance is improved continuously. Companies are scored based on their performance over the 11 dimensions on a scale from level one to level five. The maturity matrix has an approach that offers a systematic way to evaluate and guide the transition towards a circular product development by identifying a company's strengths and weaknesses. The assessment helps identify gaps and areas where improvements can be made to be able to prioritise efforts.

CSFP Quadrant Model & DfPC Toolkit

The Closing-Slowing Future-Past (CSFP) quadrant model is a framework designed to classify circular product attributes based on two key strategies: closing resource loops (e.g., through recycling) and slowing them (e.g., through durability or reuse) (Shevchenko, Yannou, et al. 2022). In addition, the model distinguishes whether a product's circularity contribution relates to past resource use—such as incorporating recycled materials during the design phase—or to its future potential, like being designed for recyclability.

The model defines four quadrants based on the combination of these dimensions. The past closing the loop part includes products that already contain recycled materials which contributes to circularity by utilising existing resources. The future closing the loop part consists of products that are designed to be recyclable by ensuring that the materials can be recovered at the end of their lifecycle. The future slowing the loop part focuses on products which are designed for reuse, repair or remanufactured and thereby extending their functional life. Lastly, the past slowing the loop part includes products that are made with reused components that allows for avoiding waste by repurposing materials that have already been in circulation.

In addition, the model identifies fifteen circular product categories, each that represents different combinations of closing and slowing loops over both past and future timeframes. The categories allow for a detailed assessment of how the products integrate circular strategies across various phases of their lifecycle. The structured method helps companies and researchers to evaluate the product's overall circularity, while at the same time identifying areas where the potential for circularity is underutilised. The CSFP quadrant model supports companies in handling material management more effectively and in addition integrating circular principles into product design.

Schevchenko with collegues have also developed an extension of the CSFP model called the Design for Product Circularity (DfPC) Toolkit (Shevchenko, Cluzel, et al. 2024). While CSFP is mainly a classification and measurement tool, DfPC expands into practical application, helping companies not just measure circularity but also improve it through better design. It includes a hierarchical calculation method, starting from individual components and parts, which then develops into a comprehensive Product Circularity Index. DfPC toolkit includes 15 guidelines for circular product design, 43 strategies for implementing circularity improvements and 225 sub-strategies offering concrete design actions.

Functionality Over Use-Time

Functionality Over Use-Time (FOUT) is a model that focus on maintaining the product's functionality across its entire lifecycle rather than just tracking the material flows (Hatzfeld et al. 2022). The FOUT model was developed in response to limitations in existing circularity indicators, which often emphasise material recirculation but overlook how well a product continues to fulfill its intended function over time. Instead of focusing solely on keeping materials in use, FOUT shifts the emphasis toward maximising the utility and value of products and materials for as long as possible.

The model represents product circularity by plotting **functionality** on the y-axis and **time in use** on the x-axis in a graph. The graphs visualisation helps in capturing how the performance of a product degrades or is retained through use, repair, reuse and recycling phases. One key principle with FOUT is that a material or product that no longer serves a function becomes waste, which in practice means that the central aim is to delay or prevent this loss for as long as possible.

This model provides a method to assess circularity based on function preservation over time, which enables more informed design decisions that emphasise reuse, durability and true value retention. In this way, FOUT works as a complement to other circularity models by adding a performance based perspective that highlights the importance of extending the useful life of products and materials as a core strategy.

3 Method

This chapter will describe the methods used to carry out this thesis. It will begin with a description of the design of the research and highlight the delimitations. The chapter will then explain the collection and analysis of the data and concludes by providing an assessment of validity.

3.1 Research Design

The chosen method followed a structured process grounded in the International Organisation for Standardisation (ISO 59010 2024; ISO 59004 2024; ISO 59020 2024), which offers internationally recognised guidance for measuring and assessing circularity performance. These process steps were selected to ensure that the resulting indicators and evaluation process followed a scientifically standardised approach and were practically applicable in a business context. Given the thesis' dual focus on product-level circularity and business relevance in a Product-as-a-Service (PaaS) model, this standards-based approach enables consistent alignment with regulatory expectations (e.g., CSRD and ESRS) while also allowing for the development of custom, decision-relevant indicators tailored to the micromobility sector. The method was therefore set up according to the following five steps:

Step 1 - **Context & Boundaries**: In which the system in focus should be identified and temporal and geographical boundaries are set. A map of the resource inflows and outflows should also be documented.

Step 2 - **Selecting Indicators**: Core circularity metrics such as recyclability, durability and resource retention as well as additional indicators like social, environmental and economic impacts should be set.

Step 3 - **Data Collection**: Collect primary and secondary data. As well as ensuring the validity of the data gathered.

Step 4 - **Perform Circularity Assessment**: Where an analysis of the alignment to existing CE principles and methods was to be done as well as an assessment of the value and impact.

Step 5 - **Perform & Monitoring** In which the results are documented and a system for periodic monitoring should be done.

The use of both qualitative (literature review, semi-structured interviews) and quantitative (material analysis, indicator evaluation) methods reflected the interdisciplinary nature of CE assessments and supported a comprehensive understanding of both technical and strategic dimensions of circularity at the case company.

3.1.1 Literature Study

The theoretical framework was built on a literature study, where the sources were found through a systematic search for literature using databases such as Scopus, Web of Science, and the Library database at the Royal Institute of Technology using the following search strings:

("model" OR "framework") AND ("Circular economy") AND ("measuring" OR "metrics" OR "indicators")

("Circular economy") AND ("Indicators") AND ("model" OR "Framework") AND ("Measuring Circularity" OR "Sustainable design" OR "Resource loops" OR "Design for Sustainability" OR "Product-Service System")

The following set of criteria was also used to help ensure that the selected literature was of sufficient quality.

- The articles should be published in high-impact and peer-reviewed journals and have good empirical data.
- The sources used in the framework should include a diversity of author's approaches on the subject, and the selection should represent a balance of conceptual frameworks with both quantitative and qualitative studies. In addition, it represents different perspectives on the topic of circularity and sustainability.

The articles that resulted were then filtered by citations and publication year (2014-2025). A time span was set in order for the models to be as closely aligned with today's ISO-standards as possible. To then decide which frameworks/models to evaluate, the following initial requirements were also set:

- The model/framework has to include and/or connect to indicators
- The model/framework has to focus on circularity on the product/meso-level

Evaluating the models was done through a set of criteria that can be viewed in Table 4.

3.1.2 Context & Boundaries

In accordance with ISO 59010 and 59020, the context and system boundaries were identified. The geographical boundary was limited to Stockholm, where the case company has its primary operational hub. Limiting the system to one city helps avoid the complications and variability that could come from analysing multiple cities and countries, each with differing regulations, user behaviours, and operational practices. The focused approach aimed to minimise the risk of inconsistent data and uncertainty in the findings.

The temporal boundary was set to year 2024, and the product of focus was the case company's version of e-scooters available on the streets 2024. For product information, such as material, components and assemblies, the Bill of Materials (BOM) was used. When that information was lacking, as in specific weight of some part components, information was drawn from earlier generation of scooters. Any deviations from the defined system boundaries, where necessary, are clearly stated and described. See figure 1 for a simplified presentation of the system boundaries.

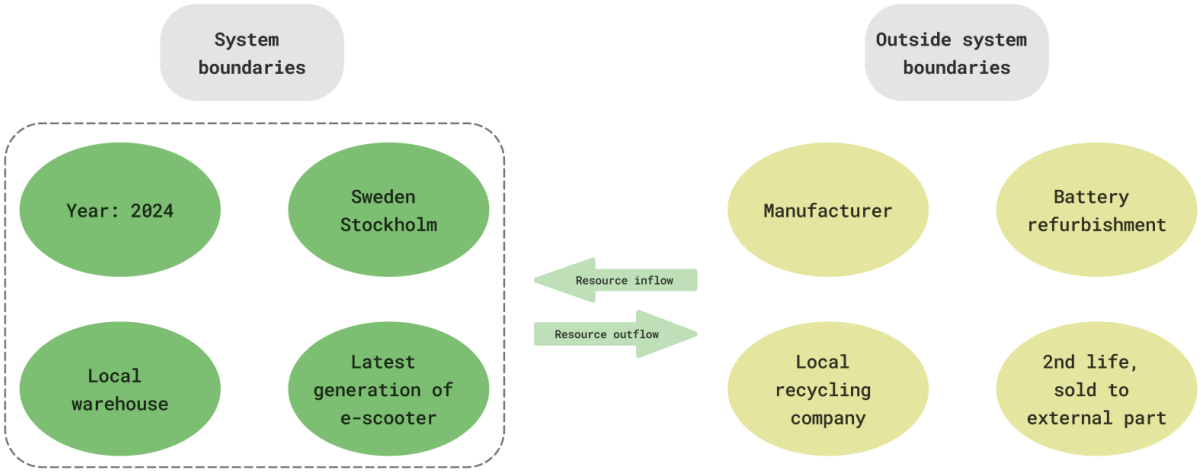


Figure 1: Visualisation of the system boundaries

The system boundaries and the "outside system boundaries" are still considered as two connected systems that interact with each other in terms of resource outflows and inflows. This is an important detail, as certain essential data, such as information connected to the design or production phase of the scooter, required collection beyond the defined system boundaries.

3.1.3 The use of AI

OpenAI's ChatGPT has been used throughout this project as grammatical support to improve the readability and language of the report.

3.2 Data Collection

3.2.1 Primary data

In addition to the literature study, semi-structured interviews were conducted. This was done to gain a deeper understanding of circularity measurement and how it can be applied on micromobility. The semi-structured format allowed for balancing predefined questions with open ended discussions, which enabled the interviewees to elaborate on key areas within their expertise.

The interviews conducted served two main purposes. First, interviews within the case company were done with the intent to understand the business model and to map out the resource inflows and outflows, in accordance to Step 1 in the research design. Some of the internal interviews were focused on identifying existing circular strategies as well as the challenges and opportunities within the case company's operations. Main interviewees in this phase were held with managers over different operation areas such as procurement, supply chain and quality assurance. These interviews aimed to provide a comprehensive understanding of the product and processes from manufacturing to operations and management in field.

Secondly, interviews outside of the company were conducted to investigate, validate and expand findings from the literature study. The focus in this phase was to expand knowledge of CE principles and circular business models. As well as gaining support in the material analysis of recycled materials. The interviews and the literature study were not strictly linear but rather parallel to build upon each other iteratively. This approach allowed for both academic and industry perspectives to be dynamically integrated throughout the research.

The interviews are presented in Table 3. "Internal" interviews are referring to the interviews held in the case company and "External" refers to interviews outside of the company. The interviewees were chosen based on their expertise and relevance, with a diverse range of perspectives. The interviews were recorded and transcribed.

Table 3: List of interview objects

Number (#)	Position	Affiliation	Duration (min)
1	Head of Supply Chain	Internal	43
2	Responsible for Procurement and Supply Chain	Internal	40
3	Vehicle Quality Assurance Engineer	Internal	25
4	Expert in Circular Business Models	External	41
5	Expert of Business Design	External	40
6	Professor in Eco Design	External	52

3.2.2 Secondary data

The secondary data used in this study included the BOM for the product in focus, along with data from the case company's internal database (2024), which covered information such as repair records, number of rides, transportation activities, downtime, and related operational metrics. Additional data was also gathered from the case company's waste management

partner and from internal files shared by key personnel when information could not be extracted directly from the databases.

3.3 Data Analysis

3.3.1 Selection Criteria for Model Evaluation

To analyse the chosen models and their applicability to this thesis case, a set of criteria was developed to evaluate each measuring model. The criteria provided a structured outline that was used to compare models with different scopes, methodologies, and applications.

These criteria guided the selection, comparison, and integration of models that support the development of circularity indicators at the product and business model levels. The aim was to ensure alignment with both theoretical frameworks and practical requirements specific to the micromobility sector and Product-as-a-Service (PaaS) business models.

Each criterion was developed to assess the capacity of a model to address one or more of the following objectives;

- Alignment of standardised circularity indicators
- Coverage of key CE strategies
- Applicability at the product or meso level
- Ability to business decision making in PaaS contexts
- Flexibility to allow development of customised indicators
- Inclusion of environmental and economic performance
- Practical considerations of scalability and data intensity
- Validation and proven practical application and strength in visualisation and communication

The list of the set criteria and their motivation can be seen in Table 4.

The chosen models were then evaluated using a Pugh Matrix, a decision-making tool used to compare alternatives against a set of criteria relative to a baseline. In this case, the matrix was built to assess each model's level of alignment with the defined criteria, using Life Cycle Assessment (LCA) as the baseline (see Table 5). The ranking was based on a qualitative interpretation of each model's scope, purpose, and documented application, as derived from the original literature and relevant case studies. Scores ranged from **-** (appears worse than baseline), **0** (equal to baseline), to **+** (appears superior to baseline). The results were then weighted to identify the models with the strongest overall alignment.

3.3.2 Selecting Indicators

Based on the weighted results from the Pugh Matrix and the ISO 59020 list of core and additional indicators - the process of selecting indicators started. The additional indicators were chosen by qualitatively assessing the most aligned CE models as well as the pre-study and interview with experts within the area. This involved identifying the gaps in the Pugh Matrix as well as making assessments on what additional indicators were relevant to the micromobility case.

The process began with a review of the core and additional indicators proposed in ISO 59020, which define circularity in terms of material inputs, outflows, and end-of-life pathways. These indicators served as a foundational layer, offering a standardised vocabulary and structure for assessing circularity across systems. Indicators were assessed in relation to the case boundaries (temporal, geographical, and organisational) to determine their applicability.

However, while the ISO 59020 framework provides a standardised approach, it is designed to be broadly applicable across industries and business models, and therefore allows for a high degree of interpretation in how indicators are selected, defined and implemented.

To supplement the ISO framework, a model evaluation was carried out using a Pugh Matrix that compared five circularity assessment models (CPDM2, DfPC, FOUT, CSFP, and PLCM) against 11 criteria, including product-level applicability, alignment with ISO standards, and support for business relevant decision making. These criteria were established based on a combination of literature and stakeholder needs. The highest scoring models in the evaluation process were then used to inform the selection of non-ISO indicators, particularly in areas not sufficiently addressed by the standard.

To address these gaps, context-specific custom indicators were developed, using a combination of insights from the model evaluation, literature, and expert interviews. These were selected based on their relevance to circularity strategies (e.g., slowing the loop) and their potential to support internal decision-making in a practical way. Only indicators that were feasible to measure with available data, or could be feasibly introduced into the case company's systems, were considered.

Table 4: List of criteria and their motivation

Criteria	Motivation
1. Product/meso level applicable	CE indicators are frequently developed at the macro scale (system/national). For this study they should be applicable to the product or meso level to evaluate micromobility systems.
2. Address slowing, closing, narrowing loops	The models used need to adress core strategies like narrowing, slowing and closing resource loops to ensure alignment with the ISO standards and the definitions of circularity defined in this thesis.
3. Aligns with ISO 59010/59020 (core & additional indicators)	Alignment with the ISO standards is crucial to ensure that the indicators meet internationally recognised standards to facilitate comparability and standardisation.
4. Business relevant / PaaS decision-making	Given the industry context, the chosen models need to include insights that supports decision-making connected to product design, maintenance and lifecycle management.
5. Supports development of additional indicators	This study aims to develop tailored indicators for the case context so the chosen models need to be adaptable to support the creation of context-relevant metrics.
6. Includes carbon impact & cost implication	Evaluating environmental and economic impacts is critical to justifying the circular strategies in a business context. Models that enable integration with LCA and cost analysis is therefore important for this analysis.
7. Data intensity and feasibility	Data availability and complexity impact the feasibility of applying the models in real business settings. This criteria concerning data intensity were therefore included to reflect data limitations.
8. Scalable/Practical for micromobility context	Models must be scalable across product types and practical for industry implementation.
9. Degree of validation	Models need to have a proven applicability or a high degree of validation to ensure the credibility of this study's findings.
10. Potential to communicate and visualise	The results of this study will also be presented and used by the collaborating company. It is therefore important that the models can be communicated clearly for usability and stakeholder engagement.

3.3.3 Material Analysis

To identify the critical components in terms of their impact on environment, the material software Granta EduPack was used. This tool enabled a structured analysis of the material

within the scooters components, covering data on carbon emissions (CO_2) and energy usage. The material analysis was done in three steps. First, all components and their material contents [4] were mapped in Granta EduPacks tool Eco Audit. For part components made of multiple materials, estimations were made by dividing the total weight of the part component proportionally among the amount of different materials-meaning that the exact proportion of each material was not known. With the help of Eco Audit, a simple Life Cycle Assessment and report was produced on the product phases material, manufacture, energy, and transportation. From this report, key components were identified based on their distinctly higher CO_2 emissions compared to the rest of the components. As a second step, two alternative theoretical scenarios were created by increasing the amount of recycled materials within the identified key components. One scenario in which all material within the key components were set to Eco Audits "typical %", and one extreme case where all material was set to a 100% recycled material rate.

The three scenarios were then compared to assess how a higher proportion of recycled content would impact CO_2 emissions. The comparison between the different cases allowed for evaluating environmental benefits as well as support in the investigation of how the choice of materials affect overall circularity.

3.4 Validation

Validation of the results was carried out in several steps and through different approaches. The findings were continuously reviewed and discussed both with internal experts at the case company as well as external experts with knowledge in the field of relevance. The results of the Material Cases were discussed and reviewed together with the Professor in Eco Design.

In addition to this, stakeholder involvement throughout the entire period of the project enabled ongoing alignment with the needs and opportunities of the case company. Regular check-ins and discussions made sure that the research remained focused and relevant as well as keeping a foundation in practical needs.

Further on, the methodology of the project was structured around the ISO 59020 standard for circularity measurement. This standard provided a systematic and standardised foundation for conducting the research which enhances the reliability and consistency of the process. Although considerable efforts were made to validate the methodology and findings, assumptions and uncertainties remain and are addressed in the analysis.

[4] All material data came from the case company's BOM

4 Results & Analysis

This section will provide the results from the model evaluation, present the indicator framework, the analysis of the material and then finally delve into the results of business alignment and the development of the business case for the Case Company.

4.1 Model Evaluation

The models were qualitatively evaluated using a Pugh Matrix, and the LCA framework was used as a baseline model to compare with [5]. The result of this evaluation can be seen Table 5.

Table 5: Pugh Matrix scoring of the chosen models, compared to LCA as baseline

Criteria	Baseline LCA	CPDM2	DfPC	CSFP Quadrant	PLCM	FOUT
1. Product/meso level applicable	0	+	+	+	+	+
2. Address slowing, closing, narrowing loops	0	+	+	+	0	+
3. Aligns with ISO 59010/59020 (core & additional indicators)	0	0	0	0	-	-
4. Business relevant / PaaS decision-making	0	+	+	0	+	0
5. Supports development of additional indicators	0	+	+	+	0	+
6. Includes carbon impact & cost implication	0	-	0	-	+	0
7. Data intensity and feasibility	0	+	0	+	0	-
8. Scalable/Practical for micromobility context	0	+	+	+	+	-
9. Degree of validation	0	+	+	+	0	0
10. Potential to communicate and visualise	0	+	+	+	+	+
Weighted total	0	7	7	6	4	1

By cross-referencing the strengths of each model with the indicator categories defined in ISO 59020, the matrix helped guide the development of the indicator framework. This process highlighted which additional indicators were supported by high-scoring models and revealed gaps where no existing indicators aligned well. These gaps informed the proposal of custom indicators tailored to the specific needs of micromobility.

The weighted analysis of the Pugh matrix also shows that the model DfPC has the strongest alignment with practical implementation needs, design adaptability, stakeholder

[5]The LCA baseline was based on the LCA ISO 14040:2006

communication, and product-level application. Additionally it identifies CPDM2 as being strong in strategic maturity evaluation and is well suited for business-oriented decision-making frameworks. CSFP provides value through classification and visualisation, while PLCM and FOUT is more specialised in or academic in scope, with limited alignment to ISO or practical feasibility.

While the Pugh matrix allowed for structured comparison, it is important to note its limitations, including subjectivity in scoring and the lack of quantitative precision. This is discussed more in section 5.5. The models that were more academic in scope, still provided guidence when choosing additional and custom indicators. DfPC and CPDM2 was useful for the business case as a tool to visualise and communicate the current state of the case company's circularity efforts, this as well as showing areas for improvements and where to focus efforts moving forward.

An assessment of the case company was made using CPDM2. The visualisation of the assessment can be seen in figure 2.

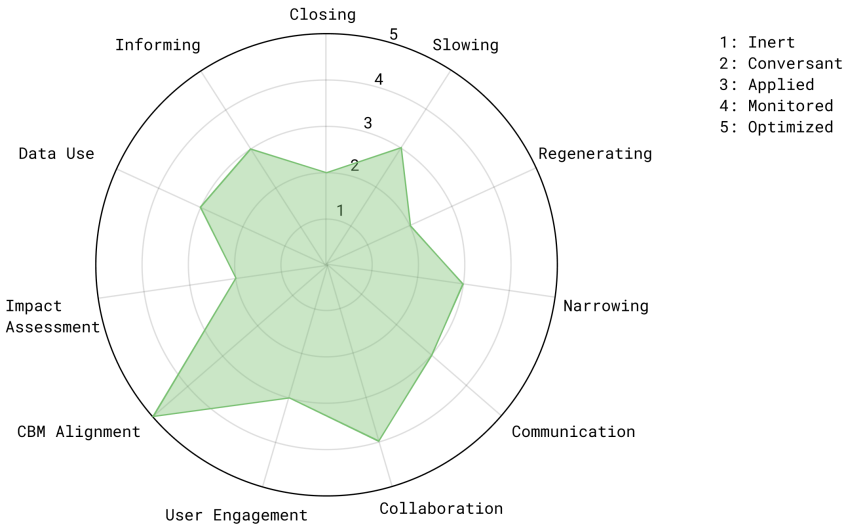


Figure 2: A Radar chart showing the case company's scoring of CPDM2

The detailed scoring of CPDM2 can be seen in appendix A.2. How the model works is described in greater detail in section 2.2.2. From this assessment it becomes clear that the case company have areas of their current circular efforts that have much room for improvement, and some areas that are well developed.

The most conspicuous areas of improvement for the case company are the dimensions: Impact Assessment, Closing and Regenerating. The CPD Impact Assessment is a part of the design process category. The dimension refers to the level of measurement that the case company has integrated. As of today the case company does not have a systematic way of measuring their circularity efforts but have recognised the need for it. Therefore they are scored as conversant for this dimension. The Closing and Regenerating are two dimensions part of the design strategies group. Regenerating means designing with renewable, non-toxic and eco-friendly materials. Closing means designing for recyclability, mono-material use, disassembly, biodegradability etc.

Due to the type of product that the case company has, it makes sense that the regenerating and closing dimensions are low. They will also be more difficult to increase in maturity since it demands research into material options and selection.

4.2 Indicator Framework and Calculations

The developed Indicator Framework is presented in Table 6. What is seen in the table is a list of ISO-Mandatory indicators but also optional, additional as well as two custom indicators. The selection of additional and custom indicators was guided by the most relevant CE measurement models as a results of the Pugh Matrix as well as the priorities and insights provided by the experts interviewed to help create as comprehensive a measurement framework as possible.

The indicators are divided into three categories, **A** (Resource Inflow), **B** (Resource Outflow), and **C** (Custom). The additional indicators in category **B** (B5 and B6) was chosen due to the support of DfPC and CPDM2 for product strategy and due to the relevance to the degradation and remanufacturing potential of micromobility products. The only optional indicator (B1) was deemed relevant due to its alignment with DfPCs design logic and PaaS business lifecycles.

> *" They ['the case company'] could look at ways to increase the utilisation rate of the scooters. Then you really have to look at what is preventing them from being used more. Is it because people aren't placing them in the right spot? Is it because they keep discharging all the time? For example"* (Expert in CBM)

Beyond the Core and Additional Indicators, it was decided that two custom indicators were to be added to the framework in order to fill clear gaps in decision relevance in short-cycle service models. It was clear from the pre-study that Utilisation Rate was an important measurement when it comes to evaluating PaaS companies. Especially in terms of connecting the circularity impact with generating value from the products. It's because of this that the other custom indicator; Time-to-Repair also becomes important for this specific case. Repairs are directly connected to circularity strategies as a part of the R-strategies but for the case company, it also affects the vehicles amount of time in field. For a PaaS company, less time in field means they are not making revenue for that period. Therefore it was argued that this would be an important indicator to include. These custom indicators do not aim to replace ISO indicators but serve as lightweight, actionable extensions for high-rotation product contexts like micromobility.

The following section presents the results of the selected indicators, along with the corresponding calculations and justifications.

A1: Average reused content of inflow

This refers to when materials and parts reach end-of-life and are reintroduced into the system through remanufacturing into new products. As no materials or components are currently remanufactured after use or at end-of-life, the value for this indicator is zero.

Fraction of input material resources that are reused in components and products: 0%

A2: Average recycled content of an inflow

This indicator is based on internal documents of the case company, for which the origin and accuracy have not been independently verified.

Fraction of input material resources that is recycled material: 35.9%

A3: Average renewable content of an inflow

Renewable material, according to the Swedish ISO 59020 (ISO 59020, p. 34), is virgin biomass that fulfills the following criteria:

- replenished at a rate equal to or greater than the rate that it is extracted;
- sourced or managed in a manner compatible with sustainable development;
- produced with regenerative practices, at a minimum sustainably grown or managed, so that the services provided by these resources and other linked resources remain available for future generations.

For the case company, fraction of input material resources that is renewable content: 0%

B1: Average lifetime of product or material relative to industry average

Indicator of time that an output resource (e.g. product) will remain in use compared to an industry average for the resource. It is important to note that this is the lifetime average within the chosen system boundaries and the time that the scooter tends to stay within the system before it moves to another city or gets sold to second life. The actual total lifetime the product can fulfill is much longer. With the latest generation, the lifetime is expected to be 15 years. However, since the calculations in this thesis is only concerned with the resource flows in and out of the system boundaries, the lifetime of the scooter is set to 4 years.

B2: Percent actual reused products and components derived from outflow

The indicator **B2** that should provide a measurement of the actual reused products and components derived from the outflow was divided into three "sub-indicators" to clarify what exactly is being measuring. No parts of the outflow goes back to the production but it does get reused in terms of repair, refurbishment and second life. The three "sub-indicators" are therefore the following:

$\mathbf{B}2_1$ - Per cent of the fleet that gets a second life

There are no exact historical data on how much each generation has been sold, since the scooters tend to be kept in use for as long as possible. But three of the oldest generations have been sold for continuous operations outside of the case company. Instead its easier to look at the percentage of a fleet that gets decommissioned. [6]. If this is estimated as being the same for each year, then after four year, only 2.48 % of the fleet gets decommissioned, the rest gets a second life outside of the system boundaries.

Percentage of the fleet that gets a second life: 97%

$\mathbf{B}2_2$ - Per cent of the scooter that can be repaired with spare parts

[6]Decomissioned = Vehicles taken out of service permanently

To calculate the percentage of the scooter that can be repaired with spare parts, the total weight of spare parts were divided by the total weight of the scooter, see equation 2.

$$\text{Repaired with spare parts} = \frac{\text{Total weight of spare parts}}{\text{Total weight of Scooter}} \tag{2}$$

This resulted in a percentage of 47.6%

B2_3 - Per cent of parts that gets refurbished

The IoT (Internet of Things - electronic unit integrated in scooter) component and battery are the only two parts with trackable data connected to refurbishment and therefore the only two components used in this calculation.

$$\text{Parts refurbished} = \frac{\text{Total weight of refurbished parts}}{\text{Total weight of Scooter}} \tag{3}$$

This resulted in a percentage of 26.2%

B3: Per cent actual recycled material derived from outflow

It is important to note that this figure does not represent the proportion the scooter being recycled, but rather the recyclable share of total waste handled by the local recycling company.

Fraction of outflow that becomes recycled material: 68.27 %

B4: Per cent actual recirculation of outflow in the biological cycle

This includes any outflow that can be biodegradable and meets the qualifying conditions for recirculation. This is 0%.

B5: Per cent designed reusability rate of the outflow

This indicator was shown to be difficult to define for the case company due to the fact that they do not own the production. While the company collaborates with the manufacturer and has some influence over the design, it does not have full control or ownership of design decisions.

Given the PaaS business model, it can be argued that the scooters are designed with a high degree of reusability, as they are intended to undergo many use cycles. However, assessing how much reusability is intentionally embedded during the early design phases remains difficult to determine.

This indicator is therefore: 0%

B6: Per cent designed recyclability rate of the outflow

This indicator differs from the percentage of material actually recycled from an outflow. Instead, it reflects the intention in the design phase, as in how well components and materials are designed to be recyclable at the product's end of use, in accordance with ISO 59020.

Recyclability based on design is therefore: 0%

C1: Time-to-Repair (TTR)

Looking at the high season during 2024, May 1 to October 1, the time-to-repair gets the following measurement:

Average maintenance time it takes for the scooter to arrive at the warehouse and then be sent out to field again: 47.91h.

C2: Utilisation rate

This indicator was also defined based on the high season, spanning from May 1 to October 1, 2024. To calculate the utilisation rate, the average daily usage per scooter was identified and then divided by the total available time of a scooter per day, according to equation 4.

$$\text{Utilisation rate} = \frac{\text{Avg Daily Trip Duration}}{\text{Available Time per day}} \text{x} 100 \tag{4}$$

The total available time is set to the hours between 7am-11pm (even though the scooters are available 24h, it's between these hours that they are generally used). Calculating this, the utilisation rate is 4.34%.

This might not seem like a significantly high number, but considering that the case company has a PaaS business model, the comparison could be made with owning an e-scooter instead of renting. The Trips per vehicle per day (TVD) reveals that on average, four people use each e-scooter per day. Compared to if only one person uses the scooter each day, this would result in a utilisation rate of around 0.89%. The utilisation rate is therefore - in this case - four times higher than private-owned e-scooters (considering that the trips lengths are the same).

Utilisation rate: 4.34%

Table 6: Classification and results of indicators for the case company

Indicator Category	Mandatory or Optional	CE Indicator	Description	Value
Resource inflows	Mandatory	**A1:** Average reused content of inflow (X)	Fraction of input material resources that are reused in components and products	0%
	Mandatory	**A2:** Average recycled content of an inflow (X)	Fraction of input material resources that is recycled material	35.9%
	Mandatory	**A3:** Average renewable content of an inflow (X)	Fraction of material resources inflow (X) that is sustainably produced renewable material.	0%
Resource outflows	Optional	**B1:** Average lifetime of product or material relative to industry average	Indicator of time that an output resource (e.g. product) will remain in use compared to an industry average for the resource	4 years
	Mandatory	**B2:** Per cent actual reused products and components derived from outflow (X)	Fraction of outflow that is reused	$\mathbf{B}2_1$ = 97% $\mathbf{B}2_2$ = 47.6% $\mathbf{B}2_3$ = 26.2%
	Mandatory	**B3:** Per cent actual recycled material derived from outflow (X)	Fraction of outflow that becomes recycled material	68.27%
	Mandatory	**B4:** Per cent actual recirculation of outflow in the biological cycle	Fraction of outflow content that is recirculated at EoL for safe return to the biosphere.	0%

	Additional	**B5:** Per cent designed reusability rate of the outflow	Reusability based on design	0%
	Additional	**B6:** Per cent designed recyclability rate of the outflow	Recyclability based on design	0%
Custom	Custom	**C1:** Time-to-Repair (TTR)	TTR = Total Repair hours per unit / Number of Repaired units	47.91 h
	Custom	**C2:** Utilisation rate	Efficiency of usage	4.34 %

4.3 Material Cases

During the material phase, which refers to the material production in the start of the material life-cycle, there were two critical component groups that showed a significant larger CO_2 impact. These two were the Frame base with 23%, and the Stem with 19% of the CO_2 impact (measured in kg CO_2) respectively of the total e-scooter, see Figure 3.

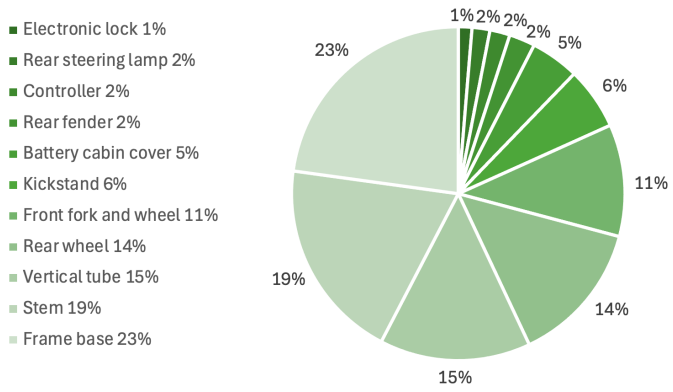


Figure 3: Shows percentage of CO_2 impact for each component group (measured in kg CO_2)

To create the two comparison cases, the material contents of these two component groups were modified: one case assumed a "typical" level of recycled content and the other assumed an extreme case where recycled content was increased to 100%. Typical % of recycled content refers to the recycle fraction within Granta Edupack Level 2 database and is the proportion of a materials current supply that originates from recycled sources, rather than from virgin raw materials. It indicates how much of the material in use today has been recovered and reprocessed.

The results from the comparisons of the baseline cases and the cases with an increased recycled content can be viewed in Figures 4, 5, 6 and 7.

As can be seen in Figure 4, despite the full substitution of virgin materials in the component group Frame, it can only reach a 43% reduction in CO_2 emissions. This number is slightly higher for the component group Stem as it can reach a 46% reduction in an extreme case, as can be seen in Figure 6. This is likely due to the Stem's higher proportion of materials that are more readily available in recycled form, such as aluminium. However, as previously noted, these represent idealised scenarios that are currently not achievable at scale.

The typical % comparison provides an indication of the potential CO_2 reduction that could be achieved if all applicable materials in the inflows were aligned with the typical recycled content levels for each respective material. As seen in Figure 5 and Figure 7 it was a 2% reduction in CO_2 emissions for the Frame and 11% for the Stem. The reason for also analysing the extreme cases is to benchmark the maximum potential benefit of recycling efforts across different component groups and to illustrate the theoretical ceiling of such strategies.

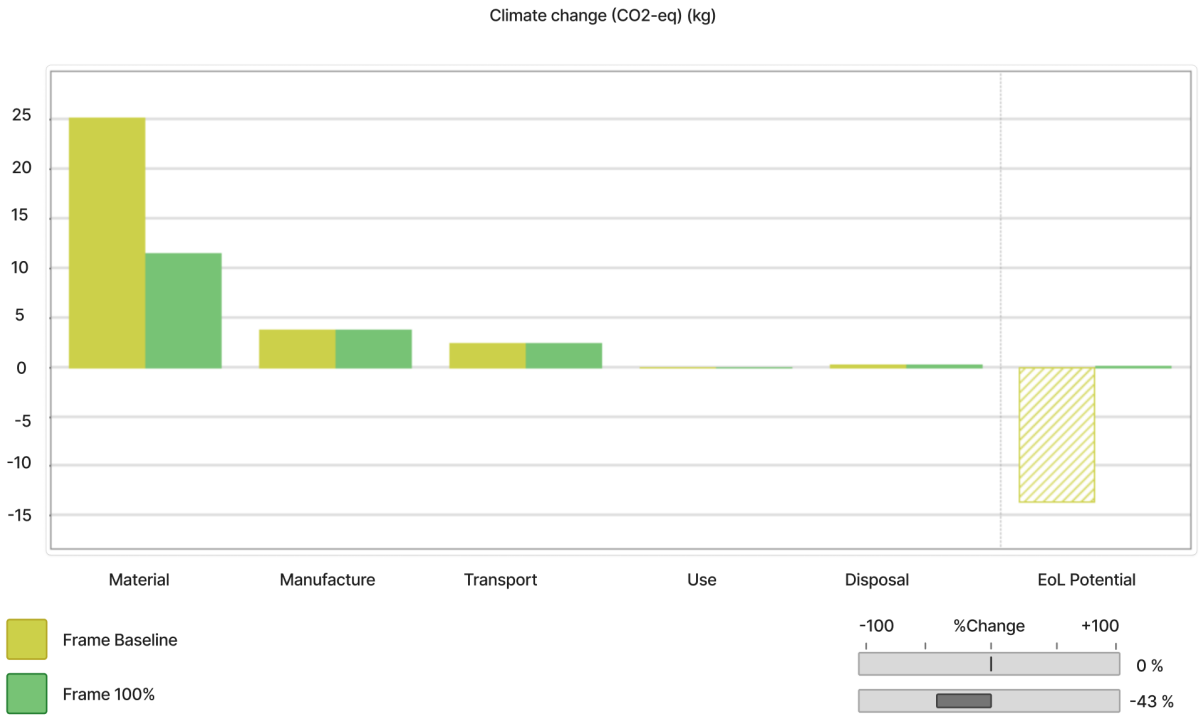


Figure 4: Frame Baseline compared to 100% recycled material

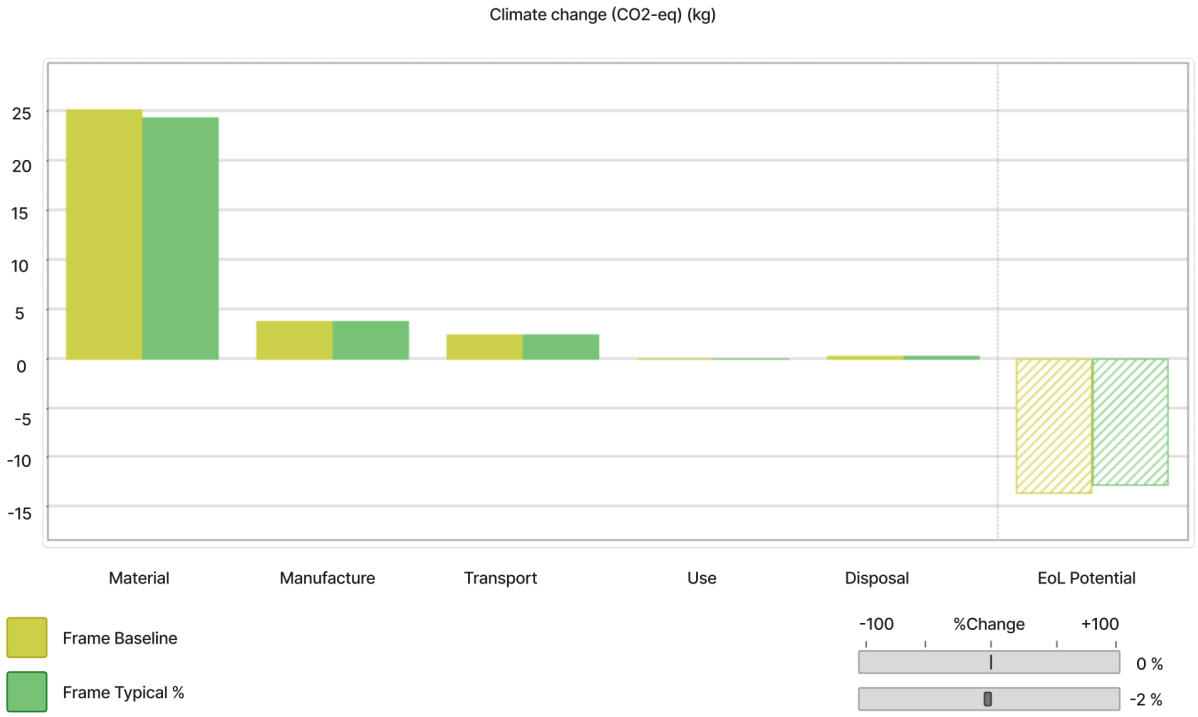


Figure 5: Frame Baseline compared to typical % recycled material

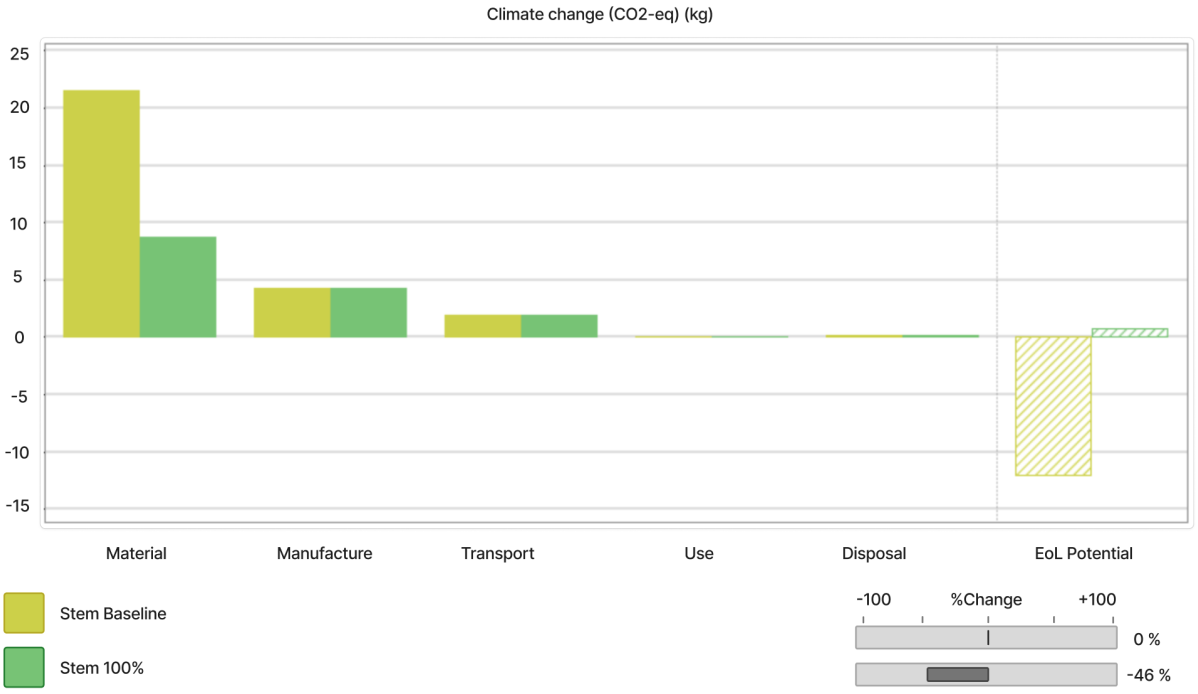


Figure 6: Stem Baseline compared to 100 % recycled material

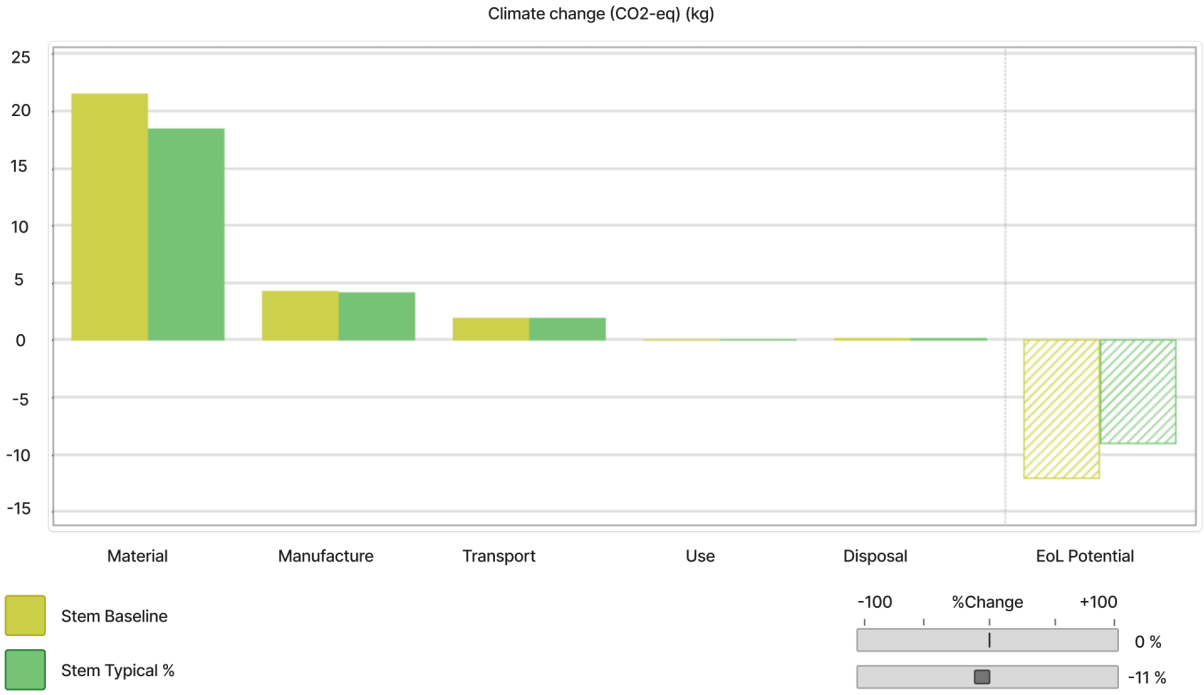


Figure 7: Stem Baseline compared to typcial % recycled material

These findings indicate that increasing the amount of recycled content does not notably reduce the footprint of the product, at least not for the components that were examined in this case. The findings of the material analysis aligns with the insights from the interviews that recycled material will have less of an impact on the overall circularity of the company compared to other resource-efficient focuses. These results are aligned with the insights provided by two of the respondents;

> *"At the same time, the impact of using recycled materials—given their particular business model—will be significantly, significantly lower than the impact of making the products last longer."* (Expert in CBM)

> *"You can see [from research] that recycling scores very low — very few have managed to make money from it. I honestly wouldn't say that circular initiatives should be based on recycling. [...]* (Business Design and CE Measurement Expert)

This means that the necessity to move the material in the components to a higher recycle rate should most likely not be considered a priority strategy for the case company to improve their environmental performance in the near future.

4.4 Business Alignment

To ensure that the circularity initiatives also are strategically aligned with the core business logic of the case company, this section explores how the developed circularity indicator framework supports business decision-making. Specifically, it assesses how circular strategies—such as increasing recycled content, improving repairability, and maximising product utilisation—translate into tangible value within the case company. The basis for the

circular business case will be presented together with the strategic recommendations and implementation of the CE indicators.

The case company's business model already inherently promotes circularity and resource efficiency. The focus has therefore been to look at aspects in their business that causes the scooters to reduce its value and what aspects that hinders them from reducing their primary resources. Something that the Expert in CBM respondent touched upon:

> *"And given that they have a high utilisation rate, you could say that they are very circular—since it takes very few resources to deliver this service over a long period of time. But then you could also say that what hinders this are various factors that reduce its value, because as soon as it can no longer generate revenue for them, it essentially no longer holds any value for them."* (Expert in CBM)

What the indicator scores and CPMD2/DfPC showed was that their weaknesses lies primarily within measurement and design which causes them to score lower in several of the dimensions connected to design processes. Here we identify several possible improvement areas. Clear strategies formulated with the intent of managing circularity will put the case company in a better position to manage their circular activities and other activities connected to sustainability.

4.4.1 Circular Business Case

The developed business case for the case company consists of sensitive information and will not be shared in its entirety in this thesis. This section will provide information on how the findings were used to align them to business strategies and possible implementations within the case company.

The business case was constructed on the basis of the analysis of the CE measurement models, indicator framework calculations, and material cases.

To communicate the case company's current situation, weaknesses and opportunities of their circularity efforts, the results of the **CPDM2** (that can be viewed in section 4.1) was presented to the case company. Then followed an overview description of necessary context and a glossary of circularity-terms that are used in the business case.

A presentation of all indicators including thorough descriptions of all calculations and origin of the input data, followed by the analysis of options, a financial assessment, stakeholder analysis, risk assessment and finally the conclusion of the recommended solutions for the case company.

The evaluation of the business case revealed that while circularity reporting is gaining importance, operational challenges and design constraints limit the case company's ability to act on many fronts simultaneously. The analysis supports a strategy prioritisation, whereby efforts to measure and manage repair processes and utilisation rates are considered more impactful than interventions focusing solely on recycled content. It was shown in the analysis made for the business case that the indicators with the highest potential for operational and financial value are those related to repairability ($B2_2$, C1), reuse ($B2_1$), and utilisation (C2). These indicators are also closely tied to fleet uptime and revenue generation.

The concept of care cycles was identified as an essential lens for understanding and improving product longevity and operational circularity. A care cycle refers to the end-to-end

process a product undergoes when transitioning through maintenance, diagnostics, repair, refurbishment, and reuse. These cycles reflect not just the technical feasibility of repairs but also the organisational efficiency and data infrastructure needed to manage them effectively. By categorising repairs into care cycles—such as fast-tracked repairs, bad warehouse repairs, and bad in-field repairs—the company can more accurately identify inefficiencies, design-related barriers, and resource losses in real time.

The integration of care cycle analysis into the circularity framework enables more nuanced decision-making. It goes beyond aggregate downtime and highlights operational feedback loops, allowing the organisation to prioritise design changes (e.g., modularity), workflow improvements, and staff training. Importantly, care cycles also support the development of predictive maintenance strategies, which align with circularity goals by minimising unnecessary part replacements and reducing total environmental impact.

4.4.2 Analysis of Options

Four core strategies - measuring circularity, increasing recycled content, enhancing designed reusability, and repair optimisation - were comparatively evaluated. The evaluation of the four strategies shows that the most impactful actions in the short-to-medium term are those focused on repair optimisation and circularity measurement.

First, the case of measuring circularity, as in accordance with the ISO family 59000, is crucial to be able to manage the circular activities. The strength within this option lies in the principle coined by Peter Drucker, “what gets measured gets managed”. By introducing metrics related to different stages of the e-scooters lifecycle, such as reuse and repair rates, the case company can start to monitor and improve these areas in a systematic manner. Measurement serves as a low barrier, high impact action that is less expensive than for example product redesign, however it can unlock substantial performance improvements. This is also a way to align sustainability initiatives with operational goals such as increasing the utilisation rate by more targeted repair, refurbishment and reuse strategies.

The developed indicator framework could serve as the groundwork for setting up clear KPIs in line with the overall strategy of the case company. The CPDM2 could continuously be worked on for a clear visualisation of the progress and further improvements that can be made in terms of circular practices. The primary cost associated with this option is internal resources for defining metrics and setting up structured tracking systems. However, most of indicators are already able to be measured.

Further on, the option of enhancing designed reusability was analysed. This option has significant potential to improve the case company’s overall circularity performance. By focusing on modular design and reusability from the early start, this option supports longer product lifespan, enabling easier refurbishment processes, and reuse of part components which are key drivers of a more circular system. As mentioned by the Expert in CBM, extending the lifespan of the e-scooters has the most substantial impact on both circularity and business models;

> *”The four strategies are: reducing material use, increasing utilisation, extending product lifespan, and enabling recirculation and regeneration. These are the four main approaches. Among them, extending lifespan and increasing utilisation are the ones that we see have the greatest impact on business models.”* (Expert in CBM)

Designed reusability directly supports this by enabling easier replacement and upgrade of components. This approach depends on collaboration with key stakeholders, particularly the manufacturer, and needs to be integrated early in the design phase. This option would also enable stronger relationships and smoothens the collaboration between the parties. Modular design not only makes it easier to reuse components, but it also simplifies disassembly, quality assurance and testing which in turn will make the refurbishment processes more optimised. Another positive outcome of this option is that in the future it could be possible to decrease the number of e-scooters purchased because it will be possible to keep them up to date just by upgrading part components. This is an alternative approach to staying competitive on the market while at the same time improving circularity. This was problematised by the Procurement and Supply chain Expert;

> *"I mean, you can keep a scooter alive just like you can with a bicycle — everything can be repaired, really. It's more that it stops being attractive to users. That's what ultimately drives us. Users choose competitors' vehicles over ours because ours are much older. That becomes a driving force for us to upgrade the fleet to something newer."*(Procurement & Supply-chain expert)

The pros of this option include improved circularity through extending product life, reducing material waste, and increasing the potential for refurbishment and reuse. However, the challenges of this lies within both design complexity, involvement and transparency from manufacturer early in the design phase, as well as most likely higher upfront development costs. Although this alternative solution has higher costs, they are balanced by long-term benefits such as lower lifecycle costs, higher utilisation and better alignment with sustainability goals.

Optimising the repair process by improving data tracking and repair friendlier design is another important step toward increasing both operational efficiency and circularity. Accessible and correct repair statistics will allow the case company to monitor trends, streamline workflows and ensure that as many e-scooters as possible remain active on the streets, minimising downtime and unnecessary storage.

The analysis of repair types revealed that nearly 14,000 repairs in Stockholm during the high season were classified as "bad" repairs. Bad repairs are described as repairs that are ineffective. Conducted either in the warehouse or in-field. These results highlight not only the need for better diagnostics but also point to a significant opportunity for cost reduction and efficiency gains. Detailed in the business case, bad repairs cost the company over €3,900 per day. Thus, repair tracking and modular redesign are key levers.

By integrating care cycle classification into repair data management, the case company can shift toward a more preventative and data-driven maintenance strategy. This aligns not only with circularity goals but also with business objectives around uptime and operational efficiency.

This option includes two main drivers, upgrading the existing data tracking system for repairs, and collaborating on the design to enable easier and faster repairs. Here, modular design also plays a key role as it enables in field repairs, reducing the need for both transport and labour. This directly supports higher utilisation rates and helps prevent out-of-service e-scooters getting piled up. Pros of this option include increased fleet availability, lower operational downtime, faster and easier repairs and better resource allocation through accurate data driven maintenance strategies. It also contributes to circularity by, once again, extending

vehicle life and reducing the environmental impact of component replacements. However, the cons include dependency on manufacturer collaboration for redesign and the need for improving tracking systems.

The cost of the repair optimisation option lies in redesigning the vehicle for modularity and internal development of tracking systems. However, the payoff lies in improved operational efficiency, lower maintenance costs and increased fleet utilisation, which all contribute positively to both financial and circular performance.

4.4.3 Financial and Strategic Implications

The business case further emphasised the economic implications of circularity strategies. For example, if design modifications lead to reduced durability (e.g., due to use of weaker recycled components), the lifespan could drop from 4 years (as baseline in this project) to 3.05 years, resulting in an estimated 24% revenue loss. This stark contrast underlines the importance of verifying material performance before implementing recycling strategies at scale.

To further support implementation, stakeholder alignment must be considered. Interviews suggested that limitations in manufacturer collaboration and product ownership reduce the feasibility of introducing modular design or upgrading key components. Municipal procurement policies and user expectations also play a role in driving fleet renewal cycles that undercut the lifespan potential.

Thus, a multi-level roadmap is recommended. First, build internal alignment through training and performance tracking (using CPDM2). Second, establish design targets for reusability and repairability in collaboration with suppliers. Third, embed indicator tracking in operational dashboards to ensure continuous feedback and improvement.

These additions strengthen the alignment between the academic evaluation and the business reality described in the accompanying business case.

5 Discussion

RQ1: In what ways can existing circular measurement models be applied for a micromobility Product-as-a-Service company?

The analysis revealed that existing circular measurement models like DfPC and CPDM2 offer practical value for evaluating circularity at the product and organisational level. These models align well with the operational and strategic needs of the case company by supporting modular assessments, identifying design and process maturity, and offering guidance for strategic decision-making. However, models such as PLCM and FOUT, while theoretically robust showed lower alignment with ISO frameworks or lacked direct business applicability in this context. Which is reasonable due to the fact that the frameworks were developed before the release of the ISO 59000 - family.

The developed indicator framework that was anchored in ISO 59020 with added custom indicators (Time-to-Repair, Utilisation Rate), demonstrated that tailoring measurement approaches to high-utilisation, shared-use systems like micromobility is both necessary and feasible. The models serve more as complementary tools than stand-alone solutions, and their combined application enabled a holistic understanding of circularity performance.

RQ2: How does increasing the recycled content of critical components within a micromobility product affect the circularity and carbon emissions of a product-as-a-service company?

The results from the material cases showed that increasing recycled content in the most high impact component groups - Stem and Controller - has a measurable but limited effect on total carbon emissions. Even under idealised conditions (100% recycled input), the reductions were modest (e.g. 17% for the Controller). These findings suggest that in the context of PaaS micromobility, prioritising strategies such as extending product lifetime, improving repairability, and maximising utilisation rates has a far greater potential to impact both environmental performance and business outcomes.

The distinction between circularity and sustainability, emphasised by Geissdoerfer et a. (2017) becomes evident here. The focus on resource loops (circularity) does not necessarily equate to broader environmental or social sustainability. For example, recycled content may appear circular, but if it requires energy-intensive processes or diverts attention from repair and reuse strategies it may not support sustainability goals. Similarly, circular design efforts without robust social systems or collaboration with suppliers may limit impact. This supports the view that circularity should be seen as a subset of sustainability, not a proxy for it.

5.1 Slowing, Closing and Narrowing Loops

The indicator framework and material case studies reinforce Bocken et al.'s (2016) distinction between slowing, closing and narrowing loops. The findings showed that slowing (through increased durability and repairability) and narrowing (higher utilisation rate) have a significantly greater impact on circular performance than closing (increasing recycled content). This supports the argument that high-level circularity is more dependent on system-level efficiency and lifecycle extension than material substitution alone. For example, increasing recycled content in critical components only led to a 17-47% reduction in emissions

in idealised scenarios, whereas higher utilisation and better repairability had both business and environmental benefits.

5.2 Rebound effects in PaaS

The case also highlights the relevance of rebound effects as outlined by Zink and Geyer (2017) and Warmington-Lundström et al. (2020). While the case company benefits from a higher utilisation rate compared to ownership-based models of almost four times higher than that of privately owned e-scooters, this increased access and convenience does not automatically translate into net environmental benefits.

Notably, the average in-system lifetime of a scooter is approximately four years, even though the technical potential is estimated to be up to 15 years for the newest generation. The decision to sell or redeploy scooters after only a few years is largely driven by competitive market dynamics and user experience expectations, rather than technical end-of-life as have been stated during the qualitative study. This reveals an important nuance in rebound logic: while PaaS systems theoretically promote longevity and efficient resource use, market preassures around eastethics, branding and user perception may shorten the actual service life of vehicles. In this case, the scooters are not discareded but often sold for second life - a circular strategy - yet their early exit from primary service reduces their value-generation potential per resource unit invested in their production.

These dynamics reflect the challenge of aesthetic obsolescence, where replacement is driven by outdated appearance or design—rather than technical failure (see Section 2.1.2). This introduces a non-technical rebound risk: systems designed for high efficiency and circularity may still under-deliver if product replacement is driven by perceived brand competitiveness rather than lifecycle performance.Thus, while the case company avoids many of the classical rebound risks like increased production volume or overconsumption, the rebound effect is embedded in branding-driven fleet refresh cycles, which can undermine circularity unless addressed through design and business model innovations (modular upgrades, in-field refurbishments, and value-based KPIs).

5.3 Limits of CE Indicators and the Role of ISO 59000

The use of ISO 59020 as a foundation for indicator selection provided a structured and standardised way to assess circularity across material flows. However, application in the case of micromobility PaaS reveals limitations in applying the ISO framework to dynamic, service-based models. While ISO 59020 provides a valuable structure for measuring circularity, its applicability across different business contexts is limited by its flexibility and need for interpretation. Beyond being a bit material-centric, the framework allows for wide variation in what is measured, how indicators are applied, and which data are included, depending on the organisation and system boundaries chosen. This makes it difficult to ensure comparability across companies or even across different internal use cases within the same organisation.

An example of this is the inclusion of the custom indicators Time-to-Repair (TTR) and Utilisation rate. These metrics were not part of the ISOs core or additional lists yet emerged as critical decision-relevant KPIs during the study. For a company whose business model depends on uptime and maximising use per vehicle, TTR has direct implications for both circularity and profitability. Similarily, Utilisation rate reflects how efficiently each product delivers value

across its service life - a factor largely overlooked by traditional CE indicators that emphasise material flows rather than use-phase performance.

These insights support critiques from Jerome et al. (2022) and Corona et al. (Corona et al. 2019) that emphasise how CE measurements often lacks alignment with real-world use, especially when frameworks rely heavily on static or predefined inputs. Thus, while ISO 59020 is a critical enabler of structured measurement, its usefulness in practice depends heavily on how indicators are selected, defined and appliead - and without complementary metrics or industry guidance, the framework risks being either too open-ended for business use or too inconsistent for comparison. In this study, integrating elements from models like DfPC and CPDM2 helped close those gaps by offering practical guidance on design maturity, reuse potential and business alignment - areas where ISO alone does not yet provide sufficient granularity.

The theory also critiques current CE measurements of overemphasising end-of-life strategies like recycling, and neglect performance during the use phase, such as durability, maintenance cycles or modular upgrades. This is particularly an issue in service-based models, where much of the circular value is generated not by avoiding waste but rather extending functional uptime and optimising use per resource unit. Further, this study found that ISO-mandated indicators such as designed recyclability or biological recirculation rate (both rated 0% in this case) offered limited business insight, either because they were not applicable to to this context (no biodegradable components) or because limitations in data (outsourced manufacturing). The thesis therefore supports the view that CE indicators and Life Cycle Assessment LCA should be treated as complementary, not competing, tools. LCA offers system-wide environmental impact visibility, while CE indicators (especially the custom) can provide operational insights and decision-making support.

In conclusion, ISO 59020 serves as a foundational layer, but in rapidly evolving, user-facing sectors like micromobility, effective measurement requires to go beyond standardisation. It should include metrics that reflect performance, design intent, and business value across the entire lifecycle — from manufacturing and first deployment to repair, reuse, and secondary use.

5.4 Implementation Challenges and Risks

While the indicator framework and material analyses point to concrete opportunities for enhancing circularity in the case company, several implementation challenges and risks need to be considered to fully realise these gains. These risks are not purely operational - they are closely tied to systemic factors highlighted in CE theory, particularly around design control, stakeholder collaboration, and the contextual nature of circularity (Geissdoerfer et al. 2017; Aguiar et al. 2022).

One of the most pressing risks is the limited design auhority held by the case company. As the company does not own the full product design, it depends on external manufacturers for making structural changes - such as increasing modularity or enabling disassembly. This aligns with the liturature that emphasises the importance of integrating CE strategies early in the design phase (Shevchenko, Yannou, et al. 2022; Aguiar et al. 2022). Without this integration, strategies like designed reusability or modular design will become difficult to implement.

Another challenge relates to the complexity of measuring circularity - particularly in

service intensive models. Even though ISO 59020 offers a clear structure its flexibility and interpretability can make indicator application inconsistent or contextually misaligned. This supports the argument made by Corona et al. (2019) and Jerome et al. (2022), who stress that indicators must be carefully tailored and iteratively tested. In this case, while the framework developed in this thesis lays a strong foundation, it must be updated and validated as new generations of vehicles, components and business needs emerge.

The realisation of circular strategies identified in this thesis depends not only on technical feasibility but also on the engagement and alignment of key stakeholders - both external and internal. These stakeholders shape the system conditions under which circularity can succeed which aligns with system level CE approaches (Bocken et al. 2016; Potting et al. 2017). The most critical external stakeholder is the manufacturer, whose collaboration is essential to enable improvements in product design - especially around modularity, disassembly and material choices. As Aguiar et al. (*DfPC toolkit* (2022)) highlights, slowing and closing loops often require design-intent and production-stage changes, which cannot be made unilaterally by service providers. To overcome this, the case company might need to move toward more formalised design partnerships or long-term strategic agreements that embed circularity goals into the procurement and design process.

Municipal stakeholders and public procurement actors are also highly relevant. As cities increasingly integrate sustainability and circularity criteria into their tenders, the ability to demonstrate measurable progress — through structured indicators — could become a competitive differentiator. This links back to the business case discussion: measurement is not just about compliance, but also about securing access to markets where circular performance is rewarded.

End-users, while not directly involved in design or operations, still play a role. Their perception of the service — including product condition, reliability, and environmental impact — can influence demand. Here, transparency around reuse, refurbishment, or second life pathways could support brand credibility and help mitigate the perceived drawbacks of ageing or reused vehicles.

Internally, cross-functional collaboration is essential. As shown in the CPDM2 assessment, current weaknesses are largely found in design maturity, strategy-setting, and coordination across the organisation. Operational teams, procurement, sustainability, and data analysts all contribute to the success of circular strategies. Without alignment and a shared set of metrics and objectives, even technically sound strategies risk poor adoption or limited impact.

While this section has discussed challenges in implementing circularity improvements, it is equally important to recognise the opportunity to use the developed indicators as a tool for managing circularity. The indicators, including Time-to-Repair and Utilization Rate, are not only relevant for reporting progress but can also serve as operational KPIs that inform product design choices, prioritise repair-related interventions, and align teams around circularity goals. Drawing on insights from the business case, these indicators can help translate abstract sustainability ambitions into tangible management actions across functions. By embedding circularity metrics in planning, follow-up, and internal communication, companies like the case firm can move from measurement to active and continuous circularity management.

5.5 Limitations

The project was subject to several limitations where one of the main constraints was related to the availability and quality of data. In some cases the data lacked sufficient detail, mainly due to limited transparency from the manufacturer. This included, amongst other things, limited access to information on subcontractors, the reasoning behind material choices, and specific design data as the case company does not fully own the product design. In addition to this, the use of material analysis software Granta EduPack included some level of generalisation in the datasets, which may not completely reflect the case company's real world processes and material sources. As a result of this, certain assumptions and estimations were necessary when specific data was unavailable. Information from earlier vehicle generations or related sources was in that case used.

Given that the nature of circularity is highly complex and multidimensional, some simplifications were required. This to keep the project within a feasible scope. As an example of this, certain process phases and material specifications were not addressed in full detail.

Although semi-structured interviews allows for exploration in depth of relevant areas, the method also introduces subjectivity as the responses may vary based on the perspectives of individuals.

Another important method to address is the *Pugh Matrix*. Even though grounded in literature, the qualitative nature introduces subjectivity in the scoring process. The evaluation relies on the authors interpretation of literature and may therefore reflect implicit biases despite efforts to ground it with a structured method. The matrix also assumes equal weighting of all criteria, which may not fully capture real-world priorities. The choice of using LCA as a baseline also frames performance in a certain way, which may influence the perceived performance of the models.

6 Conclusion

The thesis has shown that while the concept of circularity is gaining strategic importance in the micromobility sector, translating it into measurable and actionable outcomes remains complex. By applying and evaluating established circularity models against ISO standards and business needs, this study contributes a tailored measurement framework that bridges theoretical CE principles with the practical challenges of a Product-as-a-Service system.

The discussion reveals that strategies focused on slowing and narrowing resource loops—particularly those that extend product lifetime and optimise repair processes—hold far greater impact than material-focused approaches like recycling. Although recycling often dominates circularity discussions, both material analysis and expert insights confirm its limited effect in the case context. In contrast, improving repairability, modularity, and utilisation emerged as key enablers for circular and business performance.

The study also highlights the risks of rebound effects, especially in PaaS models where increased accessibility can lead to higher consumption. This underscores the need for holistic, system-aware indicators and design decisions that consider unintended consequences. The discussion also emphasises the current limitations of circularity indicators, particularly their neglect of the use phase, repairability, and functional value over time.

To move forward, companies should prioritise internal measurement capacity, adopt modular design practices, and integrate monitoring tools that are closely tied to business outcomes. The framework presented in this thesis—including both ISO and custom indicators—provides a practical foundation for doing so. While CE indicators cannot fully replace tools like LCA, they offer a structured way to assess progress, identify trade-offs, and inform design and operational decisions.

In conclusion, circularity strategies must be carefully selected and context-specific. For service models like micromobility, focusing on extending use-phase value—through product design, efficient repair, and high utilisation—is essential for both environmental impact and economic viability.

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A Appendix

A.1 Core Indicators

Indicator Category	Mandatory or Optional	CE Indicator	Summary description	Additional information
Resource inflows	Mandatory	Average reused content of inflow (X)	Fraction of input material resources that are reused in components and products	Retaining resource value
	Mandatory	Average recycled content of an inflow (X)	Fraction of input material resources that is recycled material	Add resource value
	Mandatory	Average renewable content of an inflow (X)	Fraction of material resources inflow (X) that is sustainably produced renewable material.	Add resource value
Resource outflows	Optional	Average lifetime of product or material relative to industry average	Indicator of time that an output resource (e.g. product) will remain in use compared to an industry average for the resource	Retaining resource value
	Mandatory	Per cent actual reused products and components derived from outflow (X)	Fraction of outflow that is reused	Retaining resource value
	Mandatory	Per cent actual recycled material derived from outflow (X)	Fraction of outflow that becomes recycled material	Recovering resource value
	Mandatory	Per cent actual recirculation of outflow in the biological cycle	Fraction of outflow content that is recirculated at EoL for safe return to the biosphere and meets the qualifying conditions for recirculation.	Recovers resource

Energy	Optional	Average per cent of energy consumed that is renewable	Fraction of net consumed energy that qualifies as renewable energy that qualifies as renewable energy, taking into account both energy inflows and energy outflows	
Water	Optional	Per cent water withdrawal from inflow circular sources	Per cent of annual water demand that is derived from circular sources	Maintains a circular flow of resources
	Optional	Per cent water discharged in accordance with quality requirements	Per cent (by volume) of total water withdrawn that is discharged in accordance with circularity principles	Maintains a circular flow of resources
	Optional	Ratio (on-site or internal) water reuse or recirculation	Reuse cycles of on-site water	Maintans a circular flow of resources
Economic	Optional	Material productivity	Ratio of revenue generated by total mass of all linear resource inflows	Indicates resource reduction
	Optional	Resource intensity index	Quantitative measure of economic growth versus total resource use	Indicates resource reduction

A.2 CPDM2 evaluation

	Dimensions	**Scoring**	**Scoring description**
Design strategies	Narrowing Design for resources conservation	3-Applied	Applies *Narrowing* strategies in some specific projects, focused on the design for improving production steps, aimed at reducing environmental impact, such as the design for fault diagnosis.
	Regenerating Design for regenerative systems	2-Conversant	Recognises the importance of this strategy and starts to analyse the possibility of adopting initiatives aiming to *"make clean"*, using renewable energy
	Slowing Design to slow the resource loop	3-Applied	Applies *Slowing* strategies in some specific projects, including design for product life extension and design for multiple use cycles
	Closing Design to close the resources loop	2-Conversant	Recognises the importance of this strategy and starts to analyse the possibility of adopting initiatives aimed at *closing the loop*, focusing on reincorporating post-consumption waste back into the economic cycle.

	Informing Information technologies supporting CPD	3-Applied	Applies IT in some specific projects, such as the design connected products, that advice the need for maintenance and repair and allows traceability through life cycles.
Design process	Use of data in CPD	3-Applied	Maintains a record of the results from product lifecycle analysis and starts to analyse these results aiming to improve product circularity.
	CPD Impact Assessment	2-Conversant	Recognises the importance, and has some isolated initiatives for developing indicators to identify product circularity.
	CPD and CBM Alignment	5-Optimised	Continually reviews and seeks to improve the circularity of product design aligned with the business model, examining the effects and impacts of these decisions.
	Circular User Engagement	3-Applied	Engages consumers and users in CPD
	CPD Collaboration	4-Monitored	Collaboration with stakeholders for CPD is systemised
	CPD Communication	3-Applied	Shares knowledge regarding CE and provides examples of successful applications of its principles, using case studies.

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